Int. J. Human–Computer Studies 204 (2025) 103591

Contents lists available at ScienceDirect

International Journal of Human - Computer Studies

journal homepage: www.elsevier.com/locate/ijhcs

Operationalizing selective transparency using progressive disclosure in artificial intelligence clinical diagnosis systems

Deepa Muralidhar [a,*], Rafik Belloum [b,*], Ashwin Ashok [a]

[a] Georgia State University, Atlanta, GA, USA
[b] LAMIH, CNRS, UMR 8201, Univ. Polytechnique Haut-de-France, 59313, Valenciennes, France

ARTICLE INFO

Keywords:
Transparency
LLM
Explainability
Progressive disclosure
Artificial intelligence
Human–computer interaction
CDSS

ABSTRACT

Explainable AI (XAI) is critical for clinical decision support systems (AI-CDSS) in healthcare, but current approaches often neglect the usability of explanations from a human–computer interaction (HCI) perspective. We investigate progressive disclosure as a strategy for selective transparency to provide effective explanations without overwhelming users. This paper presents a user-centered design of AI-CDSS interface prototypes that incorporate interactive explanation features (e.g., keyword highlighting of medical terms and interactive causal diagrams) and empathy-oriented nudges (e.g., supportive prompts and icons). We evaluated these prototypes through interviews with medical professionals and students, followed by a user study with general users, to assess their impact on understanding, trust, and satisfaction. Our findings suggest that progressive, on-demand disclosure of explanation details may help users manage information load and better follow the AI's reasoning process. While several interface features were well received, some elements such as affective cues like emojis elicited skepticism, particularly in clinical contexts, which underscores the importance of context-sensitive design choices.

1. Introduction

Artificial intelligence (AI) is increasingly integrated into clinical decision support systems (CDSS) to assist healthcare professionals with diagnosis and treatment recommendations. However, these AI-driven systems often operate as "black boxes", offering little insight into how they arrive at their conclusions (Peters et al., 2017; Muralidhar et al., 2023). In high-stakes domains like healthcare, this lack of transparency can erode user trust and impede the safe adoption of AI tools (Ehsan et al., 2021). Clinicians, patients, and other stakeholders therefore demand understandable explanations for an AI's suggestions (Felzmann et al., 2020). In response, the field of explainable AI (XAI) has developed numerous techniques to make AI reasoning more interpretable. Yet a key challenge remains: how to present these explanations in a user-friendly manner, providing sufficient detail to be useful without overwhelming the user.

One promising design strategy to address this challenge is *progressive disclosure*, an HCI principle that involves gradually revealing information to users as needed rather than all at once. Progressive disclosure defers complex or less immediately relevant information to secondary views, helping users focus on the most pertinent details first (Spillers, 2010). Prior research finds that the mere presence of an explanation does not automatically increase users' trust; explanations must be clear, concise, and contextually relevant to be effective (Springer and Whittaker, 2019; Muralidhar et al., 2024). In fact, overly detailed or poorly timed information can distract users and undermine their understanding of an AI system's behavior. By contrast, tailoring the timing and amount of information to the user's needs may improve comprehension and appropriate trust. While the HCI community has advocated for progressive disclosure as a means to improve transparency in interactive systems (Norval et al., 2022), there remains limited empirical evidence of its efficacy in AI-driven clinical settings.

In this work, we explore how progressive disclosure can support *selective transparency* in clinical decision support systems. We designed a prototype interface for *Docus.ai*, a commercially available AI-CDSS, that presents the AI's diagnostic reasoning in layers — from high-level summaries to detailed causal explanations — which users can reveal incrementally. The interface incorporates interactive explanation features such as keyword highlighting and interactive causal diagrams, designed to clarify how and why the system produced a given recommendation.

To further support understanding and reduce anxiety, particularly for lay users, we complemented these explanation mechanisms with empathy-oriented interface elements. These include supportive prompts, contextual icons, and affective cues intended to guide the user

* Corresponding authors.
E-mail addresses: deepa.muralidhar@gsu.edu (D. Muralidhar), rafik.belloum@uphf.fr (R. Belloum).

https://doi.org/10.1016/j.ijhcs.2025.103591
Received 17 December 2024; Received in revised form 22 July 2025; Accepted 29 July 2025
Available online 7 August 2025
1071-5819/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

and foster a more socially transparent interaction. These elements are tightly integrated into the stepwise explanation flow, allowing users to progressively uncover both informational and emotional support, adapted to the context of use. Among these, we initially hypothesized that affective cues such as emojis could help humanize the system. However, our evaluation revealed that this approach was not effective in practice: most participants, especially medical professionals, found the use of emojis inappropriate or distracting in clinical contexts. This outcome prompted us to critically reconsider the role of emotional cues in high-stakes interfaces.

We evaluated this design through a mixed-methods study including semi-structured interviews with eight physicians and eight medical students, and an online survey with fifty general users. Our findings show that progressive disclosure, when paired with carefully designed interactive features and empathetic cues, helps users manage information load, better understand the AI's recommendation and potentially fostering greater trust.

In summary, this paper contributes to the HCI and XAI literature by demonstrating how progressive disclosure can be combined with interactive explanations and social transparency cues to enhance user understanding in clinical AI systems. Through a user-centered design and empirical evaluation of a prototype AI-CDSS interface, we provide insights into how explanation strategies can be adapted to diverse user needs in sensitive, high-stakes decision contexts.

The remainder of the paper is organized as follows. Section 2 reviews relevant literature on explainable AI in healthcare and HCI approaches to explanation interfaces. Section 3 details the design of our progressive disclosure-based explanation interface prototypes and outlines the evaluation methodology, including interviews with medical experts and a user study with general users. Section 4 presents the results of the evaluation and key findings. Section 5 discusses the implications of these findings for XAI interface design. Finally, Section 6 concludes the paper and highlights directions for future work.

2. Related work

In this section, we review key literature on AI-based clinical decision support systems, explainability in healthcare, and progressive disclosure in human–AI interaction. We examine foundational and recent work to identify gaps in current approaches, particularly regarding the adaptivity, interactivity, and contextual relevance of explanations. This review situates our contribution within ongoing efforts to make AI systems more transparent and usable in clinical contexts.

2.1. AI in healthcare

Artificial intelligence (AI) promises to transform clinical decision-making by assisting healthcare professionals with diagnosis and treatment recommendations. In recent years, numerous AI-based clinical decision support systems (CDSS) have been developed, showing high performance in tasks like image interpretation and risk prediction. However, real-world adoption of AI in healthcare remains cautious. One major barrier is the *lack of transparency* of these "black-box" models, which leads clinicians to question their reliability and limits trust in AI-generated advice (Tabone and de Winter, 2023). Researchers have observed that simply deploying accurate models is insufficient for clinical acceptance—clinicians need to understand how and when to trust the system's outputs (Tonekaboni et al., 2019). In fact, low transparency is cited as a key reason many AI tools are not yet routinely used in hospitals (Tabone and de Winter, 2023). To encourage appropriate use of AI, recent efforts emphasize building *trustworthy AI* that aligns with clinical workflows and supports clinicians' existing reasoning processes (Sendak et al., 2020). Transparency is not only important for building user trust but also for meeting ethical and regulatory requirements in healthcare, which increasingly mandate explainability of AI decisions (Ghassemi et al., 2021). In summary, AI-based CDSS must be designed with human factors in mind — particularly transparency and interpretability — to bridge the gap between algorithmic performance and clinician adoption.

AI transparency in healthcare is a delicate balance. On one hand, insufficient insight into an AI's reasoning can undermine clinicians' confidence and lead to underutilization of potentially beneficial tools. On the other hand, indiscriminate trust in AI systems without understanding their limitations can be dangerous in a high-stakes domain. Indeed, overly opaque AI has been linked to errors and workflow disruptions, whereas appropriately transparent systems can improve clinicians' decision quality (Rosenfeld et al., 2019). A recent review of clinician-facing AI systems found that providing explanations generally increases doctors' trust *if* those explanations are clear, concise, and relevant, but complex or irrelevant explanations can backfire and even *decrease* trust (Rosenbacke et al., 2024). This underscores that "more" transparency is not always better; it must be the *right* transparency for the user's needs. In response, the healthcare AI community has called for human-centered design approaches that involve clinicians early in development to ensure AI tools deliver clinically meaningful information (Ehsan and Riedl, 2021). Overall, AI in healthcare motivates a strong need for tailored transparency mechanisms that engender appropriate trust and understanding among medical professionals, which provides the context for our focus on explainability and selective information disclosure in AI-CDSS.

2.2. Explainability in healthcare

The demand for *explainable AI* (XAI) in clinical settings has grown in tandem with the rise of AI decision support. Clinicians, as expert end-users, often require justification or reasoning for an AI's recommendations before integrating them into patient care (McDermott et al., 2019). Prior work in medical AI has shown that explanations can help clinicians validate model outputs against their domain knowledge, potentially catching errors or biases in the algorithm (Lundberg et al., 2018). For example, a physician using an AI diagnostic tool may want to know which patient features (symptoms, lab results) most influenced the AI's conclusion, so they can judge whether that rationale is medically sound. Accordingly, many recent studies have explored explainability specifically for healthcare AI. These include explaining machine learning predictions for ICU mortality, sepsis detection, radiology image classification, and other tasks critical to care (e.g., Carvalho et al., 2019; Tonekaboni et al., 2020). Common themes in this literature are the importance of intelligible explanations and the need to evaluate them with actual clinicians. Unlike lay users, clinicians bring extensive domain expertise and have unique expectations for explanations—they might prefer concise, clinically phrased justifications rather than technical descriptions of model internals (Belle and Papantonis, 2021). Thus, explainability techniques in healthcare must be tailored to clinical reasoning and workflows.

Despite consensus on its importance, truly effective explainability in healthcare remains an open challenge. A key issue is that many XAI methods are developed in isolation from end-user input. As noted by Tonekaboni et al. (2019), most existing explanation approaches for medical ML have not been co-designed with clinicians and lack rigorous evaluation in real clinical decision-making scenarios. This gap can lead to a mismatch between what the AI provides and what clinicians actually need to know. For instance, a heatmap highlighting pixels in an X-ray might satisfy a regulatory checkbox for "transparency" but fail to communicate anything actionable to a radiologist. Liao et al. (2020) emphasize that user needs for explanations are highly variable: different users (and usage contexts) demand different kinds and depths of explanation. In healthcare, this variability is especially pronounced—novice clinicians might need more background rationale, whereas specialists might only require confirmation of insights they suspect. Additionally, the time-critical nature of clinical decisions means explanations must not be too verbose or time-consuming. If an

AI-CDSS interrupts a clinician with lengthy, static explanations, it could hinder workflow and even patient care (Cai et al., 2019). Therefore, recent work advocates for *user-centric explainability* approaches: engaging clinicians in the design of explanation interfaces (e.g., through participatory design, as in Madumal et al., 2020) and creating adaptive explanation systems that adjust to the clinician's context and preferences (Calisto et al., 2023; de Souza Filho et al., 2024). This body of work forms the foundation for our approach, which seeks to provide explanations that are not only technically sound but also contextually appropriate and efficient for clinical users.

Another aspect highlighted in contemporary research is the distinction between explanation for *understanding* and explanation for *trust calibration*. In healthcare, the goal is not blind trust in AI, but rather calibrated trust—clinicians should trust the AI when it is correct and question it when it might be wrong (Sokol and Flach, 2020). Explanations can facilitate this by revealing the AI's reasoning process and uncertainties. For example, showing a cardiologist that an AI's prediction of heart disease risk hinges on an atypical combination of factors might prompt deeper investigation or additional tests. Conversely, if the explanation aligns with established medical knowledge, it can reinforce the clinician's confidence in the result. Recent studies have begun to measure how explanations impact clinicians' decision-making and trust. As noted, Rosenbacke et al. (2024) systematic review found mixed outcomes: in some cases explanations improved appropriate trust, but in others they did not significantly help, illustrating that explanation design and clarity are pivotal. Some have even warned that naïvely adding explanations could provide a false sense of security (Ghassemi et al., 2021)—for instance if clinicians overestimate the AI's reliability simply because it provides a rationale. These insights motivate our work to pursue more *selective and adaptive transparency*: offering the right amount of explanation at the right time, rather than a one-size-fits-all information overload. We next discuss general explanation techniques and how they can be leveraged (and extended) to achieve this balance.

2.3. Transparency and explanation techniques

Multiple strategies exist to make AI systems more transparent and explainable, ranging from inherently interpretable models to post-hoc explanation methods. In the healthcare domain, inherently interpretable models (like sparse linear models or decision trees) have been recommended for high-stakes use because their decision logic can be directly inspected (Rudin, 2019). However, such models may sacrifice accuracy on complex tasks, which is why black-box models (deep neural networks, ensemble methods) are often preferred for raw performance. To bridge this gap, a plethora of post-hoc explanation techniques has been developed. These include feature attribution methods (e.g., SHAP, LIME) that highlight which input features most influenced a particular prediction (Ribeiro et al., 2016), counterfactual explanations that describe how slight changes to an input could alter the model's prediction (Wachter et al., 2017), rule-based or example-based explanations that provide case-specific logic or similar past cases (Caruana et al., 2015; Papenmeier et al., 2022), and model cards or fact sheets that disclose an AI model's overall behavior and limitations (Mitchell et al., 2019; Sendak et al., 2020). Surveys of XAI (Guidotti et al., 2018; Arrieta et al., 2020) catalog many such techniques, More recently, Bennetot et al. (2024) provide a comprehensive tutorial on modern XAI techniques, reflecting the field's rapid progress. Many of these techniques have been applied in healthcare settings (e.g., using SHAP to interpret complex risk models for ICU patients).

While these explanation techniques advance transparency, they are typically delivered in a *static* manner. In other words, the system provides a fixed explanation (generic information output) regardless of the user's background or intent. For example, a CDSS might always display a full list of feature contributions for each prediction. This static approach has clear limitations: it can overwhelm users with unnecessary details or, conversely, provide insufficient insight if the user has deeper questions. HCI researchers have long pointed out that effective transparency is often *context-dependent* (Lim and Dey, 2009). Lim and Dey's early work on intelligibility in context-aware systems demonstrated that users have diverse *questions* about intelligent system behavior (e.g., "Why did it do X?" vs. "What if I did Y instead?") and that a system should ideally support these different queries on demand. In the AI explanation literature, similar points have been raised. Miller (2019) argues that human explanations are usually *selective*: people tend to provide only the most pertinent causes for an event, tailored to the listener's needs, rather than a complete dump of all causal factors. This suggests that AI explanations should prioritize information that is most relevant to the user's current goals or potential doubts. Moreover, providing every detail of an AI's computation could introduce cognitive overload, especially for clinicians who must interpret the explanation under time pressure. Cognitive load studies (e.g., Kaur et al., 2020) show that too much information in an explanation interface can impair user understanding and decision performance.

To address these concerns, the notion of *interactive* or *on-demand* explanations has gained traction. Instead of a single-shot explanation, the system could allow users to *query further* or reveal additional details as needed. Eiband et al. (2018) propose "transparency by design" guidelines for user interfaces, recommending that information about an AI's workings should be layered such that users can progressively dig deeper—mirroring general usability principles like Shneiderman's mantra of "overview first, details on demand" (Shneiderman, 1996). In practice, this might mean an AI-CDSS initially shows a simple justification for its recommendation (an overview), but offers controls or prompts to explore the underlying evidence or alternative reasoning if the clinician requires it. Such mechanisms prevent overwhelming the user at first glance, while still supporting thorough scrutiny when necessary. Recent human–AI interaction research reinforces this approach: Abdul et al. (2018) identify interactivity and user-control as key future directions for XAI, and Liao et al. (2020) likewise note that effective explainable AI interfaces often need to accommodate varying depths of information. In the healthcare context, an interactive explanation interface could let a doctor decide when more explanation is warranted (for instance, if a recommendation seems counterintuitive) and otherwise allow them to proceed without distraction when the AI's output aligns with expectations.

2.4. Progressive disclosure in human–AI interaction

Progressive disclosure is a well-established design principle in human–computer interaction, referring to the practice of revealing information gradually and only as needed, rather than all at once. In classical UI design, progressive disclosure improves usability by reducing information overload: interfaces present the minimum necessary information upfront and provide additional details through user interaction (e.g., clicking "More info") (Tidwell, 2010). The essence of this approach is closely related to Shneiderman's "details-on-demand" philosophy and Nielsen's usability heuristics recommending minimalist design (Shneiderman, 1996; Nielsen, 1994). In the context of intelligent systems and explainability, progressive disclosure means that an AI system might initially provide a basic explanation or indicator of confidence, with the option for the user to request more in-depth rationale if desired. This approach aligns with how people naturally seek explanations—often starting with a general answer and probing deeper only if that answer is unsatisfactory (Miller, 2019). By allowing the user to *pull* additional transparency information selectively, we avoid burdening them with irrelevant or unwanted details ("spurious explanations"), as noted by Springer and Whittaker (2020). In their seminal work on algorithmic transparency, Springer and Whittaker investigated how users react to different degrees of transparency and found that a system providing incremental, on-demand transparency can mitigate the confusion and inefficiencies caused by giving all information unconditionally. Users initially believed that more transparency would be better, but empirical tests revealed that too much low-level information, given too early, could distract users and impede their understanding of the system's overall behavior. Based on these findings, they proposed design principles for *effective transparency*, one of which is to employ progressive disclosure so that users receive simple, high-level feedback first and can delve into specifics as needed. This allows users to form a working mental model of the AI without being overwhelmed, and only later confront the AI's complexities when relevant.

Our work builds directly on the concept of progressive disclosure of explanations. Like Springer and Whittaker, we recognize the value of providing transparency "on demand" to accommodate users' varying needs. The interface we designed for the AI-CDSS uses a tiered explanation strategy: clinicians see an initial explanation snippet (e.g., the primary factor or reasoning behind a recommendation) and can request further details (such as which data influenced that factor, or how confident the system is) through additional clicks or prompts. This layered design ensures that an experienced clinician who quickly understands the AI's suggestion is not slowed down by extraneous details, while a clinician who is skeptical or curious can obtain a deeper explanation trail. The similarity to Springer and Whittaker's approach lies in this shared goal of balancing informativeness and overload by using incremental disclosure. However, our study differs in critical ways. First, the domain and user group are different: Springer and Whittaker's transparency studies were conducted in a general context of algorithmic decision-making (e.g., scenarios like algorithmic management or recommendation systems) with lay users, whereas we target a *clinical decision support* context with healthcare professionals as the primary users. This means the stakes, the nature of explanations, and the users' expertise in our work are all distinct. We leverage progressive disclosure in a high-stakes medical decision scenario, which introduces unique challenges (e.g., the need for medical validity in each explanatory step) that their general principles did not address. Second, the types of information disclosed progressively in our AI-CDSS are specialized (clinical evidence, patient-specific factors, etc.), as opposed to the more generic transparency information (like performance metrics or simple rules) studied by Springer and Whittaker. We needed to extend the progressive disclosure concept to handle domain-specific explanation content, ensuring that each layer of information is meaningful in a clinical sense (for example, disclosing a lab test result as the reason for a recommendation, and allowing further drill-down into how that lab result compares to normal ranges or past patient data). Third, our evaluation methodology diverges: rather than a lab study solely measuring users' subjective preferences and task performance with global vs. incremental transparency (as in Springer and Whittaker, 2020), we conduct an evaluation with clinicians using the system in realistic clinical decision-making tasks. Our study assesses not only whether users like the progressive disclosure interface, but also how it impacts their diagnostic decision accuracy, confidence, and workflow integration. By doing so, we provide evidence of progressive disclosure's effectiveness (or limitations) in an applied healthcare scenario, complementing the more controlled findings of Springer and Whittaker with ecological validity.

Progressive disclosure has also begun to appear in other contemporary XAI research. For instance, recent HCI work describes "selective, mutable, and dialogic" explanation systems, which allow users to select what aspect of the explanation to explore, modify certain inputs to test "what-if" scenarios, and engage in a back-and-forth inquiry with the AI (Bertrand et al., 2023). Such systems inherently rely on stepwise disclosure of information, akin to our approach. Similarly, *conversational* explanation methods have been proposed (Liao et al., 2020; Ehsan et al., 2021), in which users can ask free-form, follow-up questions after an initial static explanation, supporting a dialogue-driven form of progressive disclosure. Additionally, Bo et al. (2024) introduce the idea of *incremental explanations* for AI: they propose breaking down an explanation into a base component plus incremental details, showing that users retain understanding better when explanations are digested in pieces rather than all at once. These works reinforce the notion that a static one-layer explanation is often suboptimal; instead, an interactive, progressive approach can enhance user comprehension and satisfaction. Our contribution fits into this emerging paradigm by implementing and testing progressive disclosure in the context of an AI-CDSS, an area where, to our knowledge, this approach had not been rigorously studied before. Additionally, recent HCI design frameworks have highlighted the need to adapt interaction paradigms for human–AI interfaces, which differ fundamentally from traditional user interfaces. For example, Holzinger et al. (2022) propose the use of *personas* for AI systems to help designers anticipate user expectations, model system behavior transparently, and communicate system limitations more effectively. This approach emphasizes that human–AI interfaces should not be treated as conventional UIs, but instead require tailored design strategies that account for the system's autonomy, opacity, and evolving behavior.

In summary, prior research underscores that while AI and XAI have great potential in healthcare, existing solutions often fall short in adapting to users' needs. Traditional explanation techniques provide transparency, but usually in a fixed format that does not account for different users (or different moments in a single user's decision process). Progressive disclosure in human–AI interaction offers a promising strategy to tackle these shortcomings by tailoring the amount and depth of information to the user's current requirements. Our study situates itself at the intersection of HCI and XAI by addressing a critical gap: while many explainability methods remain generic and model-centric, clinical decision-making requires explanation strategies that are attuned to users' roles, cognitive needs, and emotional contexts. By integrating progressive disclosure into a clinical AI tool and evaluating it with actual medical practitioners, we aim to show how selective transparency can improve the usefulness and user experience of AI-CDSS. The following sections will detail our implementation of this concept and how it advances beyond prior work to fill the identified gaps.

3. Methodology

3.1. Study design

We adopted a user-centered design approach to develop and evaluate a progressive disclosure interface for an AI-driven clinical decision support system (AI-CDSS). The study was organized into two iterative phases of prototyping and feedback. In the first phase, we developed an initial prototype of a transparent user interface and conducted in-depth interviews with domain experts (practicing physicians and registered nurses). In the second phase, we refined the prototype based on the expert feedback and then gathered perspectives from a broader set of users, including medical students and end users (patients, caregivers, and other general users). This mixed-method study design allowed us to incorporate user feedback iteratively and compare insights across different stakeholder groups. Our goal was to examine whether progressively disclosing the AI system's reasoning (with interactive explanations) would enhance users' understanding and trust in the AI-CDSS, and how needs and expectations differ between expert and non-expert users.

3.2. Participants

We recruited three groups of participants representing key stakeholders of the Docus.ai system: (i) **Medical professionals** (experienced clinicians), (ii) **Medical students** (trainees in the medical field), and (iii) **General end users** (including patients, caregivers, and other non-professionals). Table 1 summarizes the demographics of these groups. In total, eight medical professionals (primarily physicians, with a few

Table 1
Stakeholder demographics: Gender and age distribution.

Stakeholder group	N	Male	Female	Age <20	20–30	30–40/>40
Doctors & Nurses	8	4	4	0	1	7
Medical Students	8	5	3	0	8	0
General Users	48	20	28	10	32	6

Table 2
Stakeholder background: AI proficiency and AI usage Patterns.

Stakeholder group	1	2	3	4	5	Used for Work (%)
Doctors & Nurses	3	1	4	0	0	12.5%
Medical Students	2	1	3	0	2	50%
General Users	4	10	23	6	5	56%

registered nurses) and eight medical students participated in the interview phases, and 50 general end users participated in the survey phase. The clinician group was largely middle-aged or older (most were over 40 years old), whereas the medical students were all in their twenties. The general end users ranged from teenagers to older adults, though the majority (approximately two-thirds) were in the 20–30 age range. We achieved a balanced gender distribution across the sample (each group had roughly equal representation of male and female participants; see Table 1). All participants had at least some familiarity with AI systems, though expertise levels varied. We asked participants to self-rate their AI proficiency on a 5-point scale (1 = beginner, 5 = expert). Clinicians mostly rated themselves around the mid-level (median rating 3), indicating moderate familiarity, while medical students were more variable (some novices and a couple with higher expertise). The general user group also clustered around intermediate proficiency (see Table 2). In terms of prior usage of AI tools, we found that a majority of general users (27 out of 50) reported using AI assistants for work-related tasks, whereas many clinicians had not yet integrated such tools into their workflow (5 of 8 clinicians indicated they had never used an AI system in practice; see Table 2). Medical students often used AI tools to support personal learning or self-improvement rather than clinical work. Participants were recruited through convenience sampling in our professional and academic networks. The medical professionals were volunteers who expressed interest in the research topic; given the challenge of recruiting busy clinicians, many had intrinsic motivation to learn about AI, and no monetary compensation was offered for this group. Medical students were recruited from a local university and were each paid a modest honorarium (USD $10) for their time. For the general user study, we distributed an online call for participants through social media and community forums; these participants were anonymous and received a small incentive (USD $5) for completing the survey. All participants provided informed consent and were aware that the study involved evaluating a prototype AI medical assistant interface.

3.3. Prototype design

We built our prototypes on the case of *Docus.ai*, a commercially available AI-CDSS. This system takes a patient's symptoms as input and produces a list of potential diagnoses and recommendations (e.g., advising further tests or actions) (Musen et al., 2001). In our study, we designed a custom user interface on top of Docus.ai's diagnostic engine to introduce transparency through progressive disclosure and interactive explanations. Two iterations of the prototype were developed: an initial design and a refined design incorporating feedback from the first iteration.

3.3.1. Initial prototype

The first prototype (see Fig. 1) introduced the concept of progressive disclosure into the AI-CDSS interface. Instead of presenting the AI's full diagnostic output on a single screen, the interface divided information into a sequence of steps that the user could navigate in order. At each step, only a manageable subset of information was shown, and the user could click "Next" to reveal more details. For example, the AI's top diagnosis suggestions were shown on an initial screen, with subsequent screens providing supporting information and explanations. We defined *transparency* in this context as the user's ability to understand *why* the AI made a certain recommendation and *how* it arrived at that conclusion. Accordingly, the prototype included two key interactive explanation features to answer the "why" and "how" questions:

- **Keyword Highlighting:** The system highlighted important symptom keywords from the user's input within the AI's textual output. These highlights indicated which input factors most strongly influenced the AI's reasoning. This design was inspired by prior work showing that highlighting relevant features can help users follow an AI's decision process (Ashktorab et al., 2019).
- **Causal Diagram (DAG):** To view a more detailed explanation, the user could click on any highlighted keyword to bring up a visual *directed acyclic graph (DAG)* illustrating the cause–effect relationships involving that symptom. The DAG showed how the symptom could lead to certain conditions or intermediate factors in the AI's reasoning chain (Fig. 1). This approach follows recommendations in explainable AI literature to use causal models for transparency (e.g., leveraging causal reasoning as per Pearl, 2019).

Through these mechanisms, the initial prototype allowed users to incrementally explore the AI's reasoning: starting from high-level results and drilling down into causal explanations only on demand. In addition to these explanation features, the interface displayed the AI's recommended next steps (such as suggested diagnostic tests or advice to seek medical care) as part of the output. In this initial design, such recommendations were presented near the beginning of the disclosure sequence, alongside the preliminary diagnosis results.

3.3.2. Refined prototype

Based on the feedback from the first-round interviews (Section 4), we refined the prototype to better align with users' expectations and address the concerns raised by clinicians. One major change was in the **information sequencing**: we re-ordered the progressive disclosure steps so that any suggested diagnostic tests or further actions from the AI were moved to the final screen of the sequence. This ensured that a user would first see the AI's diagnostic opinion and the reasoning behind it, and only later see a list of tests or referrals, thereby reducing the chance of alarming the user prematurely. We also improved the **causal diagrams** to be more clinically realistic. In the initial version, a DAG node (a symptom or finding) was sometimes linked to only a single outcome, which clinicians found too simplistic. In the refined version, the DAGs were adjusted to connect multiple related symptoms to a given potential diagnosis, illustrating a more comprehensive diagnostic pathway (incorporating suggestions from doctors to utilize existing medical causal models or "clinical pathways" for accuracy). We introduced several new interface elements in the refined prototype to enhance the system's *social transparency* and user guidance (see Fig. 2). In particular, we added what we term **nudges** — subtle UI cues designed to convey empathy and gently guide user behavior. Prior work in HCI has defined digital nudges as interface elements that influence users' decisions in a predictable way by leveraging cognitive biases (Caraban et al., 2019). In our context, the nudges were meant to make the AI assistant's responses more reassuring and context-aware for lay users. We implemented three types of nudges:

- **Emoji icon:** a small expressive icon was displayed alongside certain messages to reflect an appropriate empathetic tone (e.g., a concerned face emoji appearing next to a serious warning or a reassuring smile for positive feedback). The use of emoji was exploratory; while emojis can increase engagement in general interfaces, their effect in a serious medical context needed evaluation (Lin and Luo, 2023).

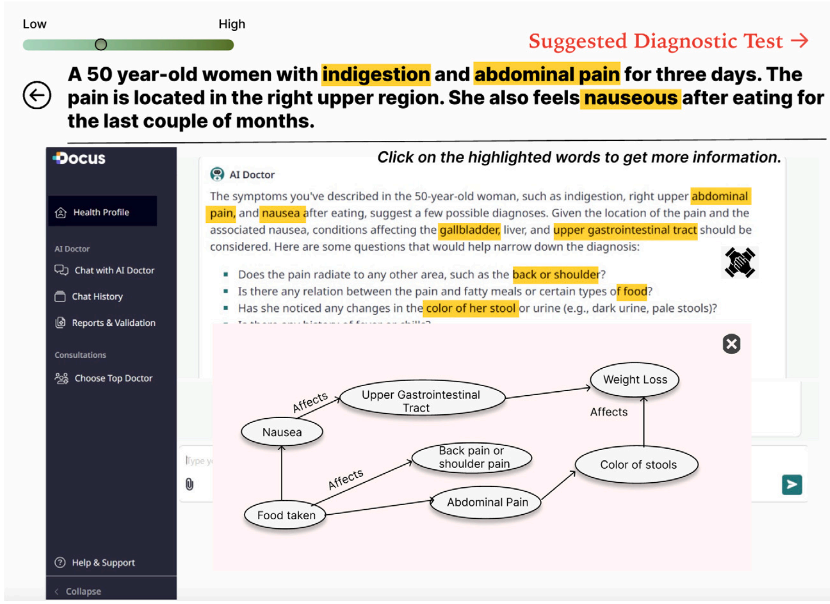

Fig. 1. The first prototype with Keyword Highlighting and Causal Diagram (DAG).

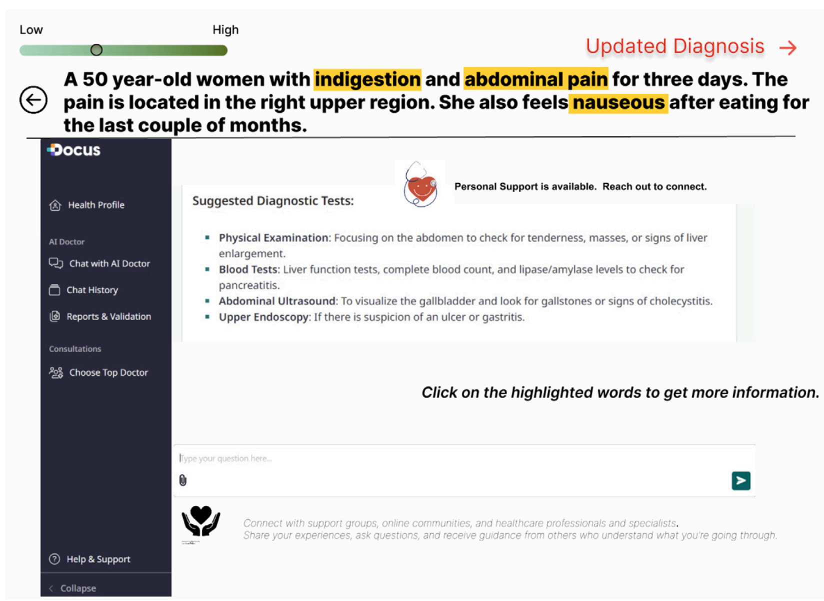

Fig. 2. The refined prototype with Nudges (Emoji icon, Contextual icon, and Trigger prompt).

- **Contextual icon:** a simple graphic icon was used as a visual "badge" for certain recommendations (for instance, a stethoscope symbol next to advice to consult a doctor, or an exclamation mark icon next to urgent recommendations). These aimed to serve as clear, professional-looking cues to draw attention without the informality of emojis.
- **Trigger prompt:** a short text prompt or link suggesting a helpful action for the user. For example, if the AI suspected a condition, the interface might show a trigger like "Learn about support groups" or "See related resources". Clicking such a trigger could provide the user with additional guidance (in the prototype, these triggers were placeholders indicating where the system could offer links or info, such as contacts of nearby clinics or support communities).

Finally, the refined prototype added a new **visual aid** in the form of a pictorial illustration for the top-ranked diagnosis. This screen showed a simplified diagram of the affected organ or system (contrasting a healthy vs. diseased state), to help users visually comprehend the medical condition. This idea stemmed from expert feedback suggesting that graphical explanations or even short videos could improve patient understanding. Fig. 3 illustrates this added explanatory image. In summary, the refined prototype maintained the core progressive disclosure and interactive explanation features of the initial version, but with improved ordering of information and additional UI elements to address user emotions and guidance. We hypothesized that these refinements would make the transparency features more palatable to clinicians and more helpful to patients, by presenting the right amount of information at the right time and in the right format.

3.4. Data collection procedure

We collected data through a combination of moderated interviews and an online survey, corresponding to the two prototype evaluation

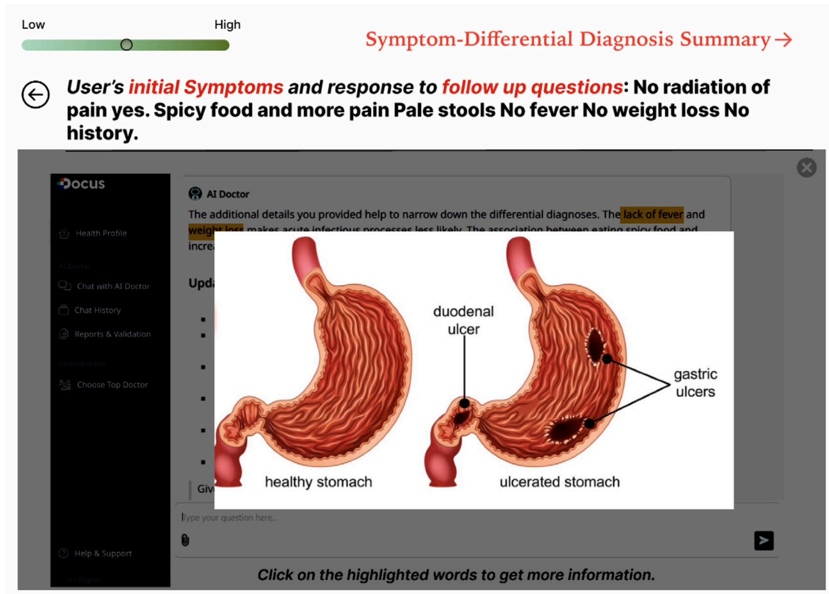

Fig. 3. The refined prototype with visual aids.

phases. In the first phase, we conducted one-on-one **semi-structured interviews** with the 8 medical professionals using the initial prototype. Each interview lasted approximately 100–120 min. At the start of each session, we introduced the context (explaining that AI-CDSS technologies are no longer confined to clinics but are now accessible to the general public) and demonstrated the existing Docus.ai interface to establish a baseline for how the AI system normally interacts with users. We then had participants engage with the prototype interface. In a think-aloud protocol, the clinician was guided through the prototype's workflow, either by the researcher sharing the screen (for remote interviews) or directly on a laptop (for in-person sessions). The participant was encouraged to verbalize their thoughts, expectations, and any confusion as they observed each step of the AI's decision-making process. After exploring the prototype, we asked the participant a series of open-ended and Likert-scale questions (via a brief on-screen questionnaire) about their perceptions of the system's transparency, clarity, and potential usefulness. These questions were designed to probe the effectiveness of specific interface features (e.g., the DAG explanations, the highlighting) and to gather suggestions for improvement. All interview sessions were audio-recorded with permission, and the researchers also took detailed notes.

Following the first set of clinician interviews, we incorporated their feedback into the refined prototype. The second phase of data collection then proceeded with two streams: additional interviews with experts (using the refined prototype) and a survey study with end users. First, we invited the clinicians for a shorter follow-up evaluation of the refined interface. Several of the original doctors and nurses participated in this follow-up (in some cases remotely). These sessions were about 45–60 min each and focused on assessing the changes: participants were shown the updated prototype (with the new features and restructured information flow) and asked for their reactions, again using a think-aloud approach and a similar questionnaire. We also recruited 8 **medical students** for interviews using the refined prototype. The student sessions followed a procedure analogous to the clinician interviews: each was 45 min long, conducted via video conferencing with screen sharing, and involved think-aloud walkthroughs of the interface followed by survey questions. All student sessions were moderated by a researcher and responses were recorded.

Finally, to capture the perspectives of general end users, we conducted an **online survey study** with the refined prototype. Participants in the survey were provided with a short video demonstrating a usage scenario of the transparent AI assistant (simulating a patient entering symptoms and the system responding through the series of progressive-disclosure screens). After watching the video, participants answered a questionnaire about their understanding, perceived transparency of the AI system, and impressions of the various interface elements. The survey contained both multiple-choice questions (including rating scales and preference selections for explanation features) and open-ended prompts for written feedback. We initially collected 30 survey responses in an asynchronous manner (participants completed the study on their own time). To ensure response quality, we then ran an additional 20 surveys in a moderated online setting: a researcher was present on a video call while the participant took the survey, available only to observe and ensure the participant followed the instructions without discussing the content. This hybrid approach helped maximize sample size while maintaining data reliability. In total we obtained 50 valid end-user responses. All participants were briefed at the beginning of the survey about the purpose of the study and what the prototype represented, and no identifying personal data was collected in the survey.

3.5. Data analysis

We employed qualitative analysis techniques for the interview data and descriptive quantitative analysis for the survey data. All interview recordings were transcribed and subjected to a thematic analysis (Braun and Clarke, 2006) to identify recurring themes and insights. Two researchers independently reviewed the transcripts and interview notes, coding segments of text that related to our research questions (e.g., comments about the usefulness of explanations, concerns about information overload, suggestions for UI improvements). The researchers then discussed and merged their codes into a set of key themes capturing participants' perceptions of transparency and the differences between stakeholder groups. Representative quotes were extracted to illustrate each theme. We label quotes in the results by participant type and a number (for example, *Doctor #3* or *Student #2*) to provide context while preserving anonymity. For the survey responses, we summarized the multiple-choice and rating-scale questions using basic descriptive statistics (frequencies and percentages). In particular, we tallied user preferences for the different explanation features (e.g., how many users found the DAG useful vs. not, which type of nudge was preferred) and visualized these differences across stakeholder groups (see Section 4). Fig. 4, for instance, shows the distribution of preferred explanation modality and preferred nudge type, respectively, for each

Fig. 4. Distribution of preferred explanation features across clinicians, medical students, and general users.

group of users. Given the sample sizes, we did not perform inferential statistical tests; instead, our analysis focused on identifying notable patterns (such as features that a clear majority liked or disliked) and corroborating these patterns with the qualitative feedback from the open-ended responses. By combining interview findings with survey trends, we developed a clearer understanding of how and why users responded to progressive disclosure and to particular transparency features.

3.6. Ethical considerations

This study was approved by the institutional review board at Georgia State University. All participants provided informed consent prior to participation. For the clinician and student interviews, we explained that the study was exploratory and that their feedback would be used to improve the design of AI systems for healthcare; participants were assured that there were no right or wrong answers to our questions. Given the sensitive nature of discussing medical decision-making, we took care to emphasize that the AI system shown was a prototype and *not* to be used for real medical advice. Participants could withdraw at any time or skip any question that made them uncomfortable. Identifiable personal information was not collected; all data were anonymized and reported in aggregate or with pseudonymous labels (as described above). We also carefully managed potential power dynamics: for instance, medical students were made to understand that their honest criticism of the system was valued and would not negatively impact any academic standing. The general end-user survey was conducted anonymously online, with a debrief provided at the end to clarify any misconceptions about the AI system's capabilities. All participants (including survey respondents) were compensated appropriately for their time as described. We mitigated risks by ensuring that none of the medical scenarios presented were likely to cause undue distress and by making support information available in case participants had any concerns after the sessions. Overall, we adhered to ethical guidelines for human-subject research and prioritized the comfort and rights of our participants throughout the study.

4. Results

Overall, our findings show that the progressive disclosure approach successfully increased users' understanding of the AI-CDSS recommendations, but the desired level and manner of transparency varied notably between clinicians, medical trainees, and lay users. In the following, we first describe broad differences in stakeholders' perceptions of the system's transparency, then present their feedback on specific explanation features (highlighting and causal diagrams) as well as empathy-oriented interface cues (nudges).

4.1. Stakeholder perspectives on transparency and information disclosure

Acceptance of AI output: As a preliminary observation, all of the interviewed physicians and nurses found the AI's diagnostic results to be generally plausible. Several doctors remarked that the primary diagnosis suggested by the system was usually reasonable and in line with what a competent doctor might consider. For example, one physician noted that the AI's output was "of an acceptable level of accuracy", indicating that the core medical competence of the system was not in question. At the same time, every clinician emphasized that the AI assistant should serve strictly as a decision support tool rather than a replacement for a human doctor. They stressed that patients should ultimately consult a physician for any serious concerns, with the AI being used only to provide preliminary guidance. This sentiment was encapsulated by a participant who said that an AI medical assistant should be designed to "support a patient's understanding, *not replace professional medical care*" (Doctor #1).

Progressive disclosure vs. information overload: We observed a clear tension between the clinicians' and non-experts' preferences regarding how much information the AI should reveal to users. On one hand, practicing clinicians were cautious about the system revealing too many details to patients or laypersons. Four out of the eight doctors explicitly argued that the AI was providing "too much transparency" to the end user, meaning that it divulged aspects of the diagnostic reasoning or medical information that could confuse or worry someone without medical training. One doctor explained that if a patient sees extensive medical details, it "is going to increase anxiety in the patient" (Doctor #2). In their view, the AI should be careful not to overwhelm patients with complex information that the patients might misinterpret. On the other hand, the *general users* in our survey largely appreciated the step-by-step disclosure of information. Many end users reported that getting the AI's reasoning in gradual, bite-sized pieces actually made them feel *more at ease*. As one survey respondent put it, seeing the information organized progressively "*lowers the anxiety*" because it keeps the user "in the loop" about what the AI is considering, thereby reducing the fear of unknown "black-box" conclusions. This contrast suggests that while incremental explanation is beneficial for understanding, there is a fine line between sufficient transparency and information overload for different stakeholders. Notably, all the clinicians acknowledged that the progressive disclosure interface, in principle, helped users follow the logic of the AI system better than a traditional one-shot explanation. Despite their concerns about patient anxiety, the doctors agreed that having the AI's reasoning broken down into sequential chunks "will improve user understanding". Likewise, medical students unanimously found the progressive, multi-step revelation of information helpful. One medical student remarked that "because [the system] then tells you the reasoning, and then moves to the next step,. . . ,it helps narrow down the problem. It is actually how medicine is practiced". (Student #1). This indicates that the concept of gradually revealing information aligns with the diagnostic reasoning process taught in medical education, where one considers findings step by step.

Need for adaptive transparency: Given the divergent perspectives above, participants suggested that the interface's transparency should be adaptable to the user's background. Several medical students, when confronted with the physicians' concerns about patients seeing the same detailed information as doctors, proposed a solution: implement a toggle or mode switch for different user types. They pointed out that professional medical resources (such as the Merck Manual) often have separate versions for professionals and consumers. By analogy, our AI-CDSS interface could tailor the depth of information shown based on whether the user is a clinician or a patient. For example, one student recommended that there should be an easy way to "regulate the information displayed based on the user's profile" — essentially, a setting to show either a simplified explanation or a detailed one. Tailoring progressive disclosure to different user roles — such as clinicians, patients, and trainees — was seen as a promising approach to prevent 'exposing patients to information they should not see' while still empowering those who want to delve deeper. In summary, the stakeholder feedback underscored the importance of *selective transparency*: the AI should disclose enough information to be clear and helpful, but also be able to hold back or simplify certain details depending on the user's level of medical knowledge and emotional state.

4.2. Feedback on explanation features: Highlighting and causal diagrams

Usefulness of highlights: Clinical and non-clinical users generally appreciated the keyword highlighting of input symptoms as a straightforward way to understand the AI's reasoning. Several end users mentioned that highlighting made the diagnosis results "less scary" because it explicitly pointed out the factors the AI was considering, which demystified the outcome. One user noted that by seeing certain words highlighted, they understood *why* those symptoms led to the AI's conclusion, and this clarity reduced their worry about the result. Clinicians likewise found the highlights useful as a quick summary of what the AI thought was important. A doctor confirmed that "Highlighting is useful – it shows the symptoms so we can come up with the diagnosis", indicating that this feature aligned with how they themselves would reason from symptoms to illness. However, the experts also urged caution: a highlighted symptom alone might be misleading if taken out of context. As one physician warned, "*That particular symptom by itself is not an indication [of diagnosis]. This may be accurate for a specific case but not for all.*" (Doctor #4). In other words, a highlight means the AI paid attention to a symptom, but it does not guarantee that symptom is definitively predictive on its own. Another experienced doctor expressed concern about whether laypeople would interpret the highlighting correctly: "*I am not sure – as a lay person I do not know if I would understand that is a causality. I would think that they are just important words.*" (Doctor #7). This suggests that while highlighting is a useful transparency mechanism, it may require proper explanation or user education to ensure non-experts grasp its meaning. Interestingly, a few participants raised situational concerns about the highlighting feature. One end user commented that the highlights felt "invasive" if one were using the system in a public setting (perhaps because the highlighted words could draw unwanted attention to sensitive symptoms on the screen). Although this was a minor issue mentioned by a single user, it points to the need to present transparency features in a user-controlled way (for instance, allowing users to toggle the visibility of highlights if privacy is a concern).

Perceptions of causal diagrams (DAGs): The interactive DAG explanations were one of the most discussed features in the interviews. Overall, medical professionals reacted positively to the concept of visual causal graphs, and medical students were particularly enthusiastic about them. One student said the DAGs were his favorite part of the interface, explaining that "*we draw diagrams a lot to explain things... when we are providers, a diagram can explain the AI's logic. I'm a big fan!*" (Student #2). Physicians noted that the DAG view helped them think of differential diagnoses they might otherwise overlook. For example, a doctor mentioned that seeing a causal graph "*makes me think of possibilities... if I missed something, it guides me to what I should consider*". At the same time, some doctors saw potential downsides if these graphs were shown to patients. One physician feared the diagram could confuse a patient or lead them to fixate on worst-case scenarios. One physician warned that such a diagram could confuse a patient and lead them to assume the worst, causing undue panic. Another doctor pointed out that our initial implementation of the DAGs was somewhat simplistic or "inaccurate" — in the first prototype, we had sometimes linked a single symptom node directly to a diagnosis node, which does not reflect the reality that diagnoses usually emerge from combinations of symptoms. This participant argued that we "need not reinvent the wheel", noting that established medical knowledge bases and clinical pathway diagrams already exist and could be repurposed to generate more accurate causal explanations. We addressed some of these concerns in the refined prototype by connecting multiple symptoms in the DAG (rather than one-to-one links), which the doctors acknowledged as an improvement, though they still desired even greater medical accuracy in these visuals. Despite these reservations, both clinicians and students agreed that the DAG tool is valuable, especially for teaching and training. The medical students felt that the causal diagrams enhanced their understanding of the case and even imagined using such a feature as a learning aid. When we asked the students how to handle the clinicians' worry about exposing complex graphs to patients, they reiterated the idea of an adaptive interface. In their view, the system could offer "professional" and "consumer" modes: in a detailed mode, a doctor or student could see the full DAG, whereas a patient mode might simplify or omit the DAG unless the user explicitly asks for it. This feedback reinforces the earlier notion that flexibility in explanation on the UI is key.

Feedback on visual aid: We also gathered feedback on the additional visual aid introduced in the refined prototype — the pictorial comparison of a healthy versus affected organ related to the top diagnosis. This feature was very well received across the board. Many participants commented that this simple diagram helped them concretely visualize what the diagnosis meant. One medical student said it was "*my favorite frame of the entire application*" (Student #3), and a physician (Doctor #3) suggested that it would be even better if the system could present such information in a short video or animation to fully engage the patient. This indicates that augmenting text explanations with intuitive visuals can significantly enhance user comprehension and engagement.

4.3. Reactions to empathy cues and nudges (emojis, icons, triggers)

The introduction of "nudges" in the interface — especially the use of emoji reactions — sparked mixed reactions from participants, with a notable divide between the medical professionals and the other groups. We summarize these reactions by type of nudge:

Emoji reactions: The vast majority of participants, across all stakeholder categories, were uncomfortable with the use of emojis in a clinical interface. Many felt that emojis made the interaction seem too casual or even frivolous for the seriousness of medical advice. For instance, one user said, "*I don't like the emoji. I don't know what it is trying to convey.*" Another commented that "*the emojis are insensitive because when dealing with a topic that's serious and you see an emoji, it feels very casual and doesn't fit the context.*" Several doctors echoed these sentiments, warning that a patient might not take an emoji icon seriously ("Patient should not take it as a joke. So keep it serious", one physician advised). A few participants suggested that if any emotive icon is used, it should be very minimal or neutral (for example, an abstract symbol rather than a cartoon face) to avoid misinterpretation. In general, the consensus was that our attempt to convey empathy via a smiling or frowning face was not effective for this domain. This feedback aligns with findings in prior work (Luger and Sellen, 2016) that overly playful elements can backfire in high-stakes settings. As a result, while the idea of humanizing the AI was appreciated in theory, the specific implementation of emojis did not resonate with users.

Icons as indicators: Compared to emojis, simple icons (such as the stethoscope or exclamation mark in our prototype) were received more favorably. Many participants hardly commented on them — which suggests that the icons were not distracting or objectionable — and a few noted that the icons served as "good visual markers" to draw attention to important messages. Some medical students mentioned that the icons made the interface look "more professional" than using expressive emoji faces. However, participants also emphasized that the choice of icon must be appropriate and self-explanatory. An icon should intuitively match the message (e.g., a warning symbol for urgent advice, a doctor's bag for a recommendation to seek medical care). When done properly, icons were seen as a subtle way to augment textual information, especially for users who might skim the interface quickly.

Trigger prompts: The trigger-style nudges (short suggestions like "Consider reaching out to a support group") generated significant discussion, particularly around their content. End users and medical students generally appreciated these prompts as potentially useful cues. Among all the nudges, triggers were the most popular type: when asked which nudge they found most helpful, the majority of both end users and students chose the trigger prompts over emojis or icons (Fig. 2). One medical student observed, "*Yes, I believe they [the nudges] are meaningful because the lack of empathy is often one of the biggest weaknesses of AI systems. But I think the nudges should match what the user inputs. For example, the system should show more empathy for more serious medical concerns.*" (Student #4). This comment highlights that triggers — and by extension any nudges — need to be context-sensitive. In our prototype, one of the triggers simply suggested looking at support resources in general, which a couple of doctors felt was too superficial. "If the AI just says 'seek out support groups' and nothing more, that's not very useful", noted one physician, adding that the system should ideally provide a direct link or specific resource. Another medical student similarly suggested, "*It would be helpful if there's a link to connect to the support groups or online communities.*" (Student #5). These responses indicate that while users appreciate gentle prompts guiding them toward help or additional information, such prompts must provide real value (actionable next steps) rather than superficial messages.

In summary, our participants appreciated the idea of the interface gently guiding users (especially through triggers) but emphasized that the tone and format of these nudges need to be handled with care. Empathetic intent alone is not enough; the way it is delivered has to feel right for the user's situation. Medical professionals tended to be skeptical of any "soft" elements like emojis, favoring a more straightforward presentation, whereas medical students and general users were more open to interface nudges as long as they were helpful and not patronizing. This divergence again underscores the need for flexible design: an ideal system might allow more empathetic enhancements for users who are comfortable with them, and a toned-down version for those who are not. Ultimately, any added UI element (emoji, icon, or trigger) should be evaluated in terms of whether it enhances the user's understanding or decision-making process. In our case, the data suggest that triggers had the most positive impact, whereas emojis should likely be eliminated or replaced with more subtle indicators in future iterations.

5. Discussion

Our study contributes empirical evidence on how progressive disclosure can be harnessed to improve transparency in AI-driven clinical systems. The findings affirm that an interactive, stepwise explanation design can help users (both experts and non-experts) better follow an AI's reasoning, echoing prior suggestions that progressive disclosure reduces cognitive overload in complex tasks (e.g., Shneiderman, 2020). At the same time, the results highlight the importance of *calibrating* transparency to the user's context—a theme consistent with the principle of "selective transparency" in human–AI interaction (Springer and Whittaker, 2018). In our interviews, clinicians expressed concern that full transparency without filters could overwhelm patients, while end users indicated that selective glimpses into the AI's process helped ease their anxiety. This implies that designers of AI-CDSS interfaces should avoid one-size-fits-all transparency. Instead, interfaces might dynamically adjust the detail of explanations (for instance, providing simpler rationale to lay users by default, with optional deeper dives for those who want more detail). This adaptive approach aligns with the concept of *social transparency* as defined by Ehsan et al. (2021)—AI systems should account for **who** the user is when deciding **what** and **how** to explain.

Our work also sheds light on which types of transparency features are most effective for different stakeholders. The causal explanation using DAGs was particularly valued by medical professionals and trainees; this resonates with the inherently causal reasoning process in medicine and confirms that visual, structured explanations can help align AI logic with human mental models (Pearl, 2019). However, we found that these graphs must be accurate and relevant to gain expert trust. In fact, our DAG visualizations were intended as illustrative aids rather than clinically validated models; several participants pointed out their simplicity, underscoring that such explanations must be aligned with medical knowledge. Integrating vetted medical knowledge (such as existing clinical pathways) into AI explanations may increase their credibility, as suggested by some doctors in our study. For lay users, the DAGs might need to be simplified or optional—our participants themselves proposed interface controls to toggle such content. On the other hand, simple highlight cues proved to be a universally appreciated feature: they require minimal expertise to understand and provide immediate insight into the AI's focus. This supports prior research where highlighting was shown to aid users in interpreting and even correcting AI outputs (Ashktorab et al., 2019). We recommend future AI-CDSS designs include highlight-based explanations, but with clear legends or tutorials for non-expert users to prevent misinterpretation.

A notable insight from our study is the mixed reception of interface "nudges" intended to convey empathy or guide behavior. The idea of making an AI assistant more socially aware (e.g., acknowledging user emotions or prompting helpful actions) is in line with human-centered AI design philosophies. Our findings indicate that users *do* desire empathy and guidance, but the form in which these are delivered is critical. Consistent with the literature on digital nudging (Caraban et al., 2019), a gentle prompt at the right moment (such as a suggestion to seek support) can positively influence user behavior without diminishing user autonomy. Trigger prompts in our prototype were well-received when they offered concrete value, reinforcing that transparency and usability extend beyond explaining the AI's decisions to also supporting users in taking meaningful next steps. In contrast, negative feedback on emojis serves as a cautionary example: even well-intentioned features can undermine credibility when they conflict with expectations for professionalism in healthcare. This finding highlights that interface 'nudges' in healthcare AI must be carefully designed: emotional or playful cues can easily cross professional boundaries and should be used judiciously. In our case, a more neutral approach (e.g., subtle wording changes or context-appropriate icons) would likely be more acceptable than cartoon emojis. More broadly, designers should consider the ethical boundaries of 'digital nudging' in medical AI tools, ensuring that any empathy-oriented features do not undermine the system's professionalism or users' trust.

The ethical framing of our interface choices also warrants discussion. Our progressive disclosure approach implicitly assumes control over the flow of information—deciding what the user sees first and what is optional. Our study also prompts the question of who controls the level of explanation detail. Allowing the system to tailor information can prevent overload, but it may also hide details some users want. Indeed, some participants suggested adding a user toggle (e.g., a "patient mode" vs. "clinician mode") to adjust explanation depth. We recommend giving users agency in how much detail to reveal, to ensure the AI is not perceived as withholding information and to accommodate individual preferences. In essence, the adaptivity of progressive disclosure should itself be transparent, with mechanisms for users to override or adjust the disclosure settings. There is a balance to strike between providing an optimal default transparency for a given user type and respecting individual differences. In our design process, we leaned towards making the AI adapt to the presumed needs of the user (e.g., showing less technical detail to presumably non-expert users), but one could argue for giving users more direct control. As an example, a future system might start in a "basic" explanation mode but allow a curious user (even if a layperson) to flip a switch to see the expert-level explanation. This would address concerns about paternalism or unnecessary withholding of information. We touched on this idea in our results, with participants explicitly calling for user-selectable modes. Designers of explainable AI should consider not only how to adjust explanations, but also how to communicate these adjustments to users and let users participate in that decision. Ensuring that the *adaptivity* in an interface is itself transparent can enhance trust—users should not feel the AI is hiding information from them, but rather that it is presenting information in a helpful way while allowing them to explore further if they choose.

Each AI system may have different capabilities for generating explanations (e.g., some might naturally output more structured rationale, others may not support it), and the progressive disclosure UI would need to adapt accordingly. Yet, the principles we explored — layering information, interactive deep dives, user-sensitive content — are general interaction strategies that could be implemented with various underlying AI models.

Limitations and future work: This study has limitations that suggest avenues for further research. First, our sample sizes — especially the domain experts — were relatively small, and participants were recruited via convenience sampling. Thus, there is a possibility of selection bias (e.g., the clinicians who volunteered might be those more interested in AI, and the general users might skew toward younger, more tech-savvy individuals). Future studies with larger and more diverse participant groups (across different hospitals, patient populations, etc.) would strengthen the generalizability of our conclusions. Second, our evaluation was based on a prototype and scenario walkthroughs rather than long-term use of a deployed system. Users' perceptions might evolve with sustained interaction or in real clinical settings. Longitudinal studies or field deployments could examine how transparency preferences change over time or impact real decision outcomes (such as adherence to AI suggestions or trust calibration). Third, while we identified clear preferences and concerns, we did not formally measure outcomes like decision quality or cognitive load; incorporating such measures would be valuable to quantify the benefits of progressive disclosure (cf. studies in other domains that found improved task performance with adaptive explanations). Additionally, our prototype relied on a specific commercial system (Docus.ai), which may limit generalizability. Future work should apply progressive disclosure with other AI-CDSS to ensure our findings hold across different systems.

Despite these limitations, our multi-stakeholder approach provided rich qualitative insights. We believe that involving both experts and end users was a strength of the study, revealing points of agreement (e.g., both groups appreciating clear explanations) as well as tension (e.g., what level of detail is appropriate for patients). Designing transparent AI for healthcare is inherently a socio-technical challenge. Our findings reinforce that effective transparency requires balancing information richness with clarity, and that this balance must be achieved in a user-centered manner. In practice, our findings suggest that AI-CDSS interfaces could begin with an 'at-a-glance' summary for quick comprehension and allow users to interactively explore more detailed reasoning when necessary—mirroring the pattern we explored. Additionally, interfaces could be designed to detect signs of user confusion or distress, either through direct feedback or physiological cues, and adapt the explanations accordingly. This approach points to a promising direction for future research on adaptive user interfaces, particularly in healthcare decision support.

6. Conclusion

We presented a user-centered design study of transparency features in an AI clinical decision support system, evaluating an interface that uses progressive disclosure of information along with interactive explanations (keyword highlights and causal diagrams) and empathetic nudges. Through two rounds of interviews and a survey encompassing clinicians, medical students, and general users, we found that these techniques, when thoughtfully implemented, can enhance users' understanding of the AI's recommendations for our study population. In particular, providing explanations in incremental steps was helpful for making the AI's reasoning more digestible. Participants across all groups acknowledged that the transparency mechanisms — especially the visual and interactive explanations — helped demystify the AI-CDSS, thereby potentially fostering greater trust. At the same time, the study underscores the need to tailor transparency to the audience: excessive detail can overwhelm laypersons, while overly simplistic or 'friendly' cues can alienate professionals. The refined prototype that emerged from our iterative design process attempted to address these concerns by reordering content and adding user-centric features. Feedback on this version was largely positive, affirming that end users feel more informed and less anxious when they can follow the AI's thought process, and that experts feel more comfortable with AI assistance when it aligns with clinical reasoning practices. Moving forward, our research suggests that effective transparency in healthcare AI systems will require adaptable interfaces that account for user role and context. Simple additions like toggleable detail levels or context-aware prompts may make a significant difference in user acceptance. We also found that simply introducing transparency features, such as explanations and nudges, is not enough; we must iteratively refine these features based on user input, as we demonstrated. In conclusion, progressive disclosure is a promising design strategy for operationalizing selective transparency: it enables AI systems to be forthcoming with information, but on the user's terms. By empowering users with the right amount of explanation at the right time, AI-CDSS can become safer, more trustworthy, and ultimately more effective collaborators in clinical decision-making.

CRediT authorship contribution statement

Deepa Muralidhar: Writing – original draft, Resources, Investigation. **Rafik Belloum:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Conceptualization. **Ashwin Ashok:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Deepa Muralidhar reports financial support was provided by Georgia State University, Kapor Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Kapor Foundation Research Fellowship. We gratefully acknowledge the support from the Kapor Foundation, which provided the funding necessary to carry out this research.

Data availability

Data will be made available on request.

References

Abdul, A., Vermeulen, J., Wang, D., Lim, B.Y., Kankanhalli, M., 2018. Trends and trajectories for explainable, accountable and intelligible systems: An HCI research agenda. In: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, http://dx.doi.org/10.1145/3173574.3174156.

Arrieta, A.B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., Herrera, F., 2020. Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. Inf. Fusion 58, 82–115. http://dx.doi.org/10.1016/j.inffus.2019.12.012.

Ashktorab, Z., Jain, M., Liao, Q.V., Weisz, J.D., 2019. Resilient chatbots: Repair strategy preferences for conversational breakdowns. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, http://dx.doi.org/10.1145/3290605.3300484.

Belle, V., Papantonis, I., 2021. Principles and practice of explainable machine learning. Front. Big Data 4, 688969. http://dx.doi.org/10.3389/fdata.2021.688969.

Bennetot, A., Donadello, I., El Qadi El Haouari, A., Dragoni, M., Frossard, T., Wagner, B., Sarranti, A., Tulli, S., Trocan, M., Chatila, R., et al., 2024. A practical tutorial on explainable AI techniques. ACM Comput. Surv. 57 (2), 1–44.

Bertrand, A., Viard, T., Belloum, R., Eagan, J.R., Maxwell, W., 2023. On selective, mutable and dialogic XAI: A review of what users say about different types of interactive explanations. In: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. pp. 1–21.

Bo, J.Y., Kumar, H., Liut, M., Anderson, A., 2024. Disclosures & disclaimers: Investigating the impact of transparency disclosures and reliability disclaimers on learner-LLM interactions. Available as preprint (SSRN, 2024).

Braun, V., Clarke, V., 2006. Using thematic analysis in psychology. Qual. Res. Psychol. 3 (2), 77–101. http://dx.doi.org/10.1191/1478088706qp063oa.

Cai, C.J., Jongejan, J., Holbrook, J., 2019. The effects of example-based explanations in a machine learning interface. In: Proceedings of the 24th International Conference on Intelligent User Interfaces. ACM, pp. 1–6. http://dx.doi.org/10.1145/3301275.3302289.

Calisto, F.M., Fernandes, J., Morais, M., Santiago, C., Abrantes, J.M., Nunes, N., Nascimento, J.C., 2023. Assertiveness-based agent communication for a personalized medicine on medical imaging diagnosis. In: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. pp. 1–20.

Caraban, A.K., Karapanos, E., Gonçalves, D., Campos, P.F., 2019. 23 ways to nudge: A review of technology-mediated nudging in human-computer interaction. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems.

Caruana, R., Lou, Y., Gehrke, J., Koch, P., Sturm, M., Elhadad, N., 2015. Intelligible models for healthcare: Predicting pneumonia risk without shortcuts. In: Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, pp. 1721–1730. http://dx.doi.org/10.1145/2783258.2788613.

Carvalho, D.V., Pereira, E.M., Cardoso, J.S., 2019. Machine learning interpretability: A survey on methods and metrics. Electronics 8 (8), 832. http://dx.doi.org/10.3390/electronics8080832.

de Souza Filho, J.C., Belloum, R., de Oliveira, K.M., 2024. Where are we and where can we go on the road to reliance-aware explainable user interfaces? In: 2024 IEEE Symposium on Visual Languages and Human-Centric Computing. VL/HCC, IEEE, pp. 282–288.

Ehsan, U., Liao, Q.V., Muller, M., Riedl, M.O., Weisz, J.D., 2021. Expanding explainability: Towards social transparency in AI systems. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems.

Ehsan, U., Riedl, M.O., 2021. Explainability pitfalls: Beyond dark patterns in explainable AI. Patterns 5 (6), 100119, Preprint available at arXiv:2109.12480 (2021).

Eiband, M., Krause, H.C., Buss, M., Hussmann, A., Schmidt, H., 2018. Bringing transparency design into practice: User interface guidelines for explainable AI. In: Proceedings of the 23rd International Conference on Intelligent User Interfaces. ACM, New York, NY, USA, http://dx.doi.org/10.1145/3172944.3172961.

Felzmann, H., Fosch Villaronga, E., Lutz, C., Tamò Larrieux, A., 2020. Towards transparency by design for artificial intelligence. Sci. Eng. Ethics.

Ghassemi, M., Oakden-Rayner, L., Beam, A.L., 2021. The false hope of current approaches to explainable artificial intelligence in health care. Lancet Digit. Heal. 3 (11), e745–e750. http://dx.doi.org/10.1016/S2589-7500(21)00208-9.

Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D., 2018. A survey of methods for explaining black box models. ACM Comput. Surv. 51 (5), 93. http://dx.doi.org/10.1145/3236009.

Holzinger, A., Kargl, M., Kipperer, B., Regitnig, P., Plass, M., Müller, H., 2022. Personas for artificial intelligence (AI) an open source toolbox. IEEE Access 10, 23732–23747.

Kaur, H., Nori, H., Jenkins, S., Caruana, R., Wallach, H., Wortman Vaughan, J., 2020. Interpreting interpretability: understanding data scientists' use of interpretability tools for machine learning. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. ACM, pp. 1–14. http://dx.doi.org/10.1145/3313831.3376219.

Liao, Q.V., Gruen, D., Miller, S., 2020. Questioning the AI: Informing design for explainable AI user experiences. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, http://dx.doi.org/10.1145/3313831.3376590.

Lim, B.Y., Dey, A.K., 2009. Designing intelligible context-aware applications. In: Proceedings of the 11th International Conference on Ubiquitous Computing. UbiComp, ACM, pp. 193–202. http://dx.doi.org/10.1145/1620545.1620576.

Lin, T.S., Luo, Y., 2023. Health persuasion through emoji: How emoji interact with information source to predict health behaviors in COVID-19 situation. SSM - Popul. Heal. 21, 101433. http://dx.doi.org/10.1016/j.ssmph.2023.101433.

Luger, E., Sellen, A., 2016. "Like having a really bad pa": The gulf between user expectation and experience of conversational agents. In: Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems. ACM, pp. 5286–5297. http://dx.doi.org/10.1145/2858036.2858288.

Lundberg, S.M., Nair, B., Vavilala, M.S., Horibe, M., Eisses, M.J., Adams, T., Liston, D.E., Low, D.K., Newman, S., Kim, J.J., Lee, S., 2018. Explainable machine-learning predictions for the prevention of hypoxaemia during surgery. Nat. Biomed. Eng. 2 (10), 749–760. http://dx.doi.org/10.1038/s41551-018-0304-0.

Madumal, P., Miller, T., Sonenberg, L., Vetere, F., 2020. Explainable reinforcement learning through a causal lens. In: Proceedings of the 34th AAAI Conference on Artificial Intelligence. pp. 2493–2500.

McDermott, M.B.A., Wang, S., Marinsek, N., Ranganath, R., Ghassemi, M., Foschini, L., 2019. Reproducibility in machine learning for health research: Still a ways to go. In: Proceedings of the Machine Learning for Healthcare Conference. arXiv:1907.01463.

Miller, T., 2019. Explanation in artificial intelligence: Insights from the social sciences. Artificial Intelligence 267, 1–38. http://dx.doi.org/10.1016/j.artint.2018.07.007.

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I.D., Gebru, T., 2019. Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency. pp. 220–229. http://dx.doi.org/10.1145/3287560.3287596.

Muralidhar, D., Belloum, R., De Oliveira, K.M., Ashok, A., 2023. Elements that influence transparency in artificial intelligent systems-a survey. In: IFIP Conference on Human-Computer Interaction. Springer, pp. 349–358.

Muralidhar, D., Belloum, R., de Oliveira, K.M., Ashok, A., Mohammad, P.B., 2024. The effect of progressive disclosure in the transparency of large language models. In: International Conference on Computer-Human Interaction Research and Applications. Springer, pp. 269–288.

Musen, M.A., Middleton, B., Greenes, R.A., 2001. Clinical decision support systems. URL https://api.semanticscholar.org/CorpusID:10864092.

Nielsen, J., 1994. Usability Engineering. Morgan Kaufmann.

Norval, C., Cornelius, K., Cobbe, J., Singh, J., 2022. Disclosure by design: Designing information disclosures to support meaningful transparency and accountability. In: Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency. Association for Computing Machinery, http://dx.doi.org/10.1145/3531146.3533133.

Papenmeier, A., Kern, D., Englebienne, G., Seifert, C., 2022. It's complicated: The relationship between user trust, model accuracy and explanations in AI. ACM Trans. Computer-Hum. Interact. 29 (4), 1–33. http://dx.doi.org/10.1145/3495013.

Peters, J., Janzing, D., Schölkopf, B., 2017. MIT Press.

Ribeiro, M.T., Singh, S., Guestrin, C., 2016. "Why should I trust you?" explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, New York, NY, USA, pp. 1135–1144. http://dx.doi.org/10.1145/2939672.2939778.

Rosenbacke, R., et al., 2024. Heuristics and errors in XAI-augmented clinical decision-making: Moving beyond algorithmic appreciation and aversion. In: Proceedings of the 32nd European Conference on Information Systems. ECIS 2024, Kristiansand, Norway, Complete Research Paper.

Rosenfeld, A., Ben-Assuli, O., Sarne, D., 2019. A review of trust in healthcare: Can trust in AI replace trust in physicians? Comput. Hum. Behav. 93, 366–373. http://dx.doi.org/10.1016/j.chb.2018.12.016.

Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat. Mach. Intell. 1 (5), 206–215. http://dx.doi.org/10.1038/s42256-019-0048-x.

Sendak, M.P., Gao, M., Brajer, N., Balu, S., 2020. Presenting machine learning model information to clinical end users with model facts labels. Npj Digit. Med. 3, 41. http://dx.doi.org/10.1038/s41746-020-0253-3.

Shneiderman, B., 1996. The eyes have it: A task by data type taxonomy for information visualizations. In: Proceedings of the 1996 IEEE Symposium on Visual Languages. IEEE, pp. 336–343. http://dx.doi.org/10.1109/VL.1996.545307.

Shneiderman, B., 2020. Bridging the gap between ethics and practice: Guidelines for reliable, safe, and trustworthy human-centered AI systems.

Sokol, K., Flach, P., 2020. Explainability fact sheets: A framework for systematic assessment of explainable approaches. In: Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society. ACM, pp. 56–66. http://dx.doi.org/10.1145/3375627.3375830.

Spillers, F., 2010. Progressive disclosure.

Springer, A., Whittaker, S., 2018. Progressive disclosure: Designing for effective transparency.

Springer, A., Whittaker, S., 2019. Making transparency clear. In: Algorithmic Transparency for Emerging Technologies Workshop.

Springer, A., Whittaker, S., 2020. Progressive disclosure: When, why, and how do users want algorithmic transparency information? ACM Trans. Interact. Intell. Syst. 10 (4), http://dx.doi.org/10.1145/3374218, Article 41 (Dec 2020), 32 pages.

Tabone, W., de Winter, J.C.F., 2023. Using ChatGPT for human–computer interaction research: a primer. R. Soc. Open Sci. 10 (9), 231053. http://dx.doi.org/10.1098/rsos.231053.

Tidwell, J., 2010. Designing Interfaces, second ed. O'Reilly Media, Sebastopol, CA, ISBN: 978-1449379704.

Tonekaboni, S., Joshi, S., McCradden, M.D., Goldenberg, A., 2019. What clinicians want: Contextualizing explainable machine learning for clinical end use. In: Proceedings of the 4th Machine Learning for Healthcare Conference, vol. 106, PMLR, pp. 359–380.

Tonekaboni, S., McCradden, M.D., Dimakopoulos, M., Xiao, R., Lee, J., Khan, C.A., Goldenberg, A., 2020. The role of explainability in creating trustworthy artificial intelligence for health care: a comprehensive survey. arXiv:2009.xxxxx.

Wachter, S., Mittelstadt, B., Russell, C., 2017. Counterfactual explanations without opening the black box: Automated decisions and the GDPR. Harv. JL Tech. 31, 841.